

Scott Okuma

Technical Program Leadership | Hardware/Software System Integration

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Summary

Systems architect and engineering leader with 14+ years across spacecraft, embedded computing, and RF systems; responsible for spacecraft software program planning and delivery since 2020. Built Umbra's software organization, managed 12 direct reports, and coordinated with approximately 20 electrical and mechanical engineers. Translates mission needs into delivery plans, staffing forecasts, and executive progress reports. Delivered software across three spacecraft blocks successfully deployed to orbit.

Professional Experience

Umbra Space

Spacecraft Systems Architect | Goleta, CA | March 2025 to Present

Director of Spacecraft Software | Santa Barbara, CA | August 2021 to March 2025

Flight Software Lead | Santa Barbara, CA | June 2019 to September 2021

Selected accomplishments across Umbra roles

- Owned spacecraft software planning and delivery since 2020, translating mission needs into epics, features, and stories aligned with company OKRs and milestones; reported progress to the VP of Space Systems.
- Built the software department from the ground up; hired and managed 12 direct reports and established development processes, test infrastructure, and delivery practices.
- Developed Jira-based capacity forecasts and headcount proposals that informed a six-month schedule extension under funding constraints; led the team to deliver against the revised milestone.
- Led software delivery across flight computers, payload transceivers, GNC, S-band radios, and subsystem firmware, including remote updates over limited radio contacts. Delivered three spacecraft blocks to orbit with no mission losses attributable to flight software.
- Coordinated hardware/software interfaces with approximately 20 electrical and mechanical engineers; clarified CONOPs, integration responsibilities, and team-approved test plans to prepare for hardware delivery while software development progressed through simulation.
- Directed firmware adaptation for a magnetometer/sun-sensor component substitution caused by a parts shortage; defined API changes, assigned implementation, and specified verification. Replacement hardware passed initial checkout over two days: one electrical and one firmware.
- Aligned electrical and software teams on reusing existing hardware for a prototype, balancing delivery speed and engineering capacity against processor RTOS support and power constraints.
- Led the transition from bare-metal software to embedded Linux across firmware, FPGA, kernel development, and GNC model integration; reviewed processor schematics and PCB interfaces to support system integration.
- Established nightly CI/CD and software test infrastructure; developed environmental test plans covering execution and pre-/post-test functional validation for thermal-vacuum, shock, and vibration verification.

RF, Hardware, and Test Engineering

Electrical/Firmware Engineer

Raytheon - Goleta, CA | September 2017 to June 2019

- Designed digital firmware for RF system controllers using VHDL/Verilog and Vivado; customized Linux kernels and developed embedded systems with PetaLinux and C/C++.
- Conducted hardware checkout, schematic analysis, and DDR3 implementation for electronic modules, using oscilloscopes, spectrum analyzers, and network analyzers.

Electrical Engineer

NAWCWD - Point Mugu, CA | January 2015 to September 2017

- Developed software and performed RF hardware repair, layout, radar calibration, and test planning for radar systems.
- Created a DRFM jammer using an FPGA, RF transceiver, and user interface; programmed in Python, C++, and MATLAB for signal processing and simulation.

Electrical/RF Engineer

Computer Sciences Corporation - Point Mugu, CA | January 2012 to January 2015

- Developed automated test procedures for GaN RF amplifiers using LabVIEW and GPIB; built custom test fixtures and collaborated with technicians to troubleshoot systems.
- Verified equipment thermal limits using environmental chambers and data logging; analyzed test results using Octave.

University Projects

CSUN CubeSat1 - Satellite Communications Team Member

Jet Propulsion Laboratory / CSU Northridge | January 2014 to December 2014

Developed radio communication test procedures for electrostatic-sensitive devices and bit-error-rate, link-budget, and data-volume analyses for mission planning.

NOAA Radio Weather Receiver Team Member

CSU Northridge | February 2014 to June 2014

Designed an RF amplifier for a superheterodyne receiver and developed a software-defined radio using MATLAB/Simulink.

Integration and Program Capabilities

Program delivery: Milestone planning; Jira capacity forecasting; headcount proposals; executive reporting; cross-team dependencies; hiring and mentoring; integration readiness.

System integration: Hardware/software architecture; avionics interfaces; schematic and PCB review; ARM and Zynq platforms; FPGA/processor interfaces; embedded Linux; GNC model integration.

Test and verification: Environmental test planning; thermal-vacuum, shock, and vibration verification; automated RF testing; LabVIEW/GPIB; oscilloscopes; spectrum and network analyzers.

RF and software: RF system controllers; S-band radios; software-defined radio; C/C++; Python; VHDL/Verilog; MATLAB/Simulink; PetaLinux; Yocto; Vivado/Vitis; Git/GitLab.

Education and Training

Bachelor of Science in Electrical Engineering | California State University Northridge | 2012-2014

Xilinx Embedded Design with PetaLinux Tools | November 2018